

Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this .tex file directly.

For each question (that applies), replace the “Type your response here” text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace **ONLY** the “Type your response here” text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the \question command or any other lines in this document.

You can \input this .tex file right before \end{document} of your main file or compile it as a stand-alone document. Check the instructions on your conference’s website to see if you will be asked to provide this checklist with your paper or separately.

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **NA – MaskShift does not introduce a new AI method; the benchmark protocol and mask generators are described in the main paper and supplement.**
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes – The paper separates benchmark/theory claims, empirical evidence, negative diagnostics, and limitations.**
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes – Related work and dataset/mechanism cards identify the missing-data forecasting and TSF context needed to replicate the study.**

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **yes – The paper includes a theoretical risk-identity note and rank-reversal condition.**

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **yes – The matched-rate mask-shift setting, clean-target protocol, and benchmark scope are stated explicitly.**
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **partial – The formal risk identity and rank-reversal condition are stated; empirical claims are tied to audited tables rather than theorem statements.**
- 2.4. Proofs of all novel claims are included (yes/partial/no) **partial – The risk identity is elementary and explained; full formal proof expansion is not needed for the benchmark contribution.**
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **yes – The paper gives intuition for mechanism-specific Bayes predictors and model-dependent risk shifts.**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **yes – Missing-data and forecasting references are cited in the paper and audit materials.**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **yes – The theoretical warning is tested empirically through matched-rate mechanism-shift experiments.**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **yes – Negative and contrastive experiments are included, including typed-head failure and S4M robustness.**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes – The study uses Weather, Electricity, Traffic, and AirConvection.**

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **yes – Weather/Electricity provide positive evidence; Traffic/AirConvection are retained as mixed boundary datasets.**
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **NA – No new dataset is introduced.**
- 3.4. All novel datasets introduced in this paper will be

made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **NA – No new dataset is introduced.**

- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) **partial – The datasets are standard public TSF benchmarks; final public-release notes should preserve upstream source/license metadata.**
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) **yes – The datasets are public benchmark datasets used through local benchmark paths.**
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) **NA – No non-public dataset is used.**

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) **yes – The paper is primarily computational.**

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) **partial – Script configs and JSON summaries record the reduced protocols and seed offsets; the final appendix/checklist should carry the exact values.**
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) **yes – Mask generation, preprocessing, and evaluation scripts are included in the MaskShift package.**
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) **yes – Training, evaluation, statistical analysis, and table-generation scripts are included.**
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) **yes – The code can be released with the anonymous artifact bundle and public repository on publication.**
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) **NA – No new forecasting backbone is introduced; benchmark code is documented where non-obvious.**

- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) **yes – Seed offsets and multi-seed summaries are recorded in M10-M14 and the paper.**

- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) **partial – Software/runtime constraints and local reduced protocols are recorded; final submission fields should add exact hardware inventory if requested.**

- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) **yes – MSE, MAE, sMAPE, eta-squared, Kendall tau, corrected severity metrics, and uncertainty summaries are described.**

- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) **yes – The paper reports the number of seed offsets and gate-seed counts for each core result.**

- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) **yes – The analysis includes CIs, hierarchical bootstrap summaries, FDR/statistical tests, and mixed/negative boundary cases.**

- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) **yes – The package uses BH-FDR, ANOVA/Kruskal checks, and bootstrap uncertainty where appropriate.**

- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) **partial – Final model/protocol values are serialized in JSON summaries and scripts; the official form can cite those artifacts.**